AIR QUALITY, COST OF LIVING, AND CRIME ACROSS U.S. COUNTIES

by

Austin Yoo

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Committee:

_____ Dr. Kent Miller, Thesis Director

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Fairfax, VA

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Director: Dr. Kent Miller, Professor
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Dedication

I dedicate this work to my family, peers, and professors, whose unwavering support and belief in me made this journey possible. To my parents, for their endless encouragement; to my friends and professors for their support of my studies; and to my future self, who dared to ask the difficult questions and find answers in the data.

Acknowledgments

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First, I thank my advisor, Dr. Kent Miller, whose guidance, insight, and patience shaped this research from topic inception to final deliverable. His critical feedback and encouragement improved the direction of this work in meaningful ways.

I am grateful to the faculty and staff of the Computational and Data Sciences Department at George Mason University for providing the resources, seminars, and collaborative environment that made this project possible.

I also thank the compilers of the public quality-of-life dataset used in this study, and the U.S. government agencies whose records it is built from.

Lastly, I thank my family for believing in me, for tolerating my long hours, and for reminding me why this work matters.

Contents

Dedication iii

Acknowledgments iv

Abstract ix

1 Introduction 1

1.1 Research Questions 1

1.2 Contribution and Scope 1

2 Prior Research 3

2.1 Conceptual Framework 3

2.2 Population Density and Crime Scaling 3

2.3 Collective Efficacy and Social Organization 4

2.4 Environmental Stress, Heat, and Aggression 4

2.5 Air Pollution and Violent Crime 4

2.6 Lead, Water Pollution, and Developmental Effects 4

2.7 Land Use, Guardianship, and Policing 5

2.8 Socioeconomic Inequality and Cumulative Environmental Risk 5

2.9 Measurement, Identification, and Inference 5

3 Data 6

3.1 Source 6

3.2 Level of Measurement 6

3.3 Construction of the Analysis Sample 7

3.4 Data Limitations 8

4 Hypotheses and Methods 9

4.1 Hypotheses 9

4.2 Models 9

5 Results **11**

5.1 Why the First Version Overstated Significance 11

5.2 Main Estimates 11

5.3 Robustness 15

6 Conclusion **17**

6.1 Answers to the Research Questions 17

6.2 Lesson on Method 18

6.3 Limitations 18

6.4 Future Work 19

A Software **20**

Bibliography **28**

Biography **32**

List of Tables

3.1 Variables and the level at which they vary. 7
3.2 Descriptive statistics, analysis sample of counties. 8

5.1 Air quality and crime: the same bivariate slope under different units and standard errors. Slope is crimes per 1,000 residents per percentage point of good-air days. 11
5.2 County regressions of crime per 1,000 residents. Standard errors clustered by state in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Model (5) drops the District of Columbia and Utah, which have one county each in the sample. . 12
5.3 Robustness of the air-quality coefficient in Model 4. Standard errors clustered by state. 16

List of Figures

5.1 Crime and air quality across states. Each point is a state; crime is the population-weighted average over its counties, and point size reflects population. The line is the OLS fit. 13
5.2 Crime and cost of living across counties. Grey points are counties; blue points are the mean crime rate in 20 equal-size groups of counties sorted by cost of living, with 95% confidence intervals. 14
5.3 Standardized coefficients with 95% confidence intervals. Model 4 has no state fixed effects; Model 5 compares counties within the same state, so air quality (a state-level variable) drops out. 15

Abstract

AIR QUALITY, COST OF LIVING, AND CRIME ACROSS U.S. COUNTIES

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People often choose where to live based on crime, cost of living, and environmental quality. This thesis asks which of these place characteristics go along with higher crime across U.S. counties. I use a public quality-of-life dataset that combines 2016 reported crime per 1,000 residents with air quality, drinking-water violations, cost of living, median income, unemployment, and population for 2,892 counties in 49 states and the District of Columbia. A first version of this project ran the regressions on 87,457 public-school rows. Because every school in a county shares the county's crime rate, and every county in a state shares the state's air-quality value, those rows are not independent, and the conventional standard error of the air-quality slope was about 15 times too small. I therefore re-estimate the models at the county level with standard errors clustered by state, add controls for population size, urbanization, income, and unemployment, and check the results at the state level.

Three findings stand out. First, cost of living has no association with crime on its own (slope -0.33 crimes per 1,000 per \$10,000, $p = 0.62$); the significant result in the first version came from the inflated sample. Second, better air quality goes with less crime at first, but the association does not hold up: it halves once the controls are added (one standard deviation of air quality goes with about one fewer crime per 1,000, or 5% of the mean, $p = 0.03$), and it falls by another 60% and is no longer significant ($p = 0.36$) when counties are compared only with others in the same Census region. Air quality mostly stands in for broad geography, such as the dry, sunny West. Third, the features most strongly associated with crime are median income (negative) and population size and urbanization (positive). Unemployment and drinking-water violations show no association. The data are cross-sectional, and crime is measured in 2016 while income and population are measured in 2022, so all results are associations, not causal effects.

Chapter 1

Introduction

Crime is not spread evenly across cities and regions. Researchers have offered many explanations for why some places have more of it than others, including economic disadvantage, weak institutions, social norms, and the built environment. More recently, air and water pollution have been added to the list next to population density and neighborhood structure. Pollution may affect crime directly, through its effects on the brain, and indirectly, through stress, disorder, and weaker ties between neighbors [1, 2, 3, 4, 5].

1.1 Research Questions

This project was motivated by a practical question. People who move often compare places by crime rate, cost of living, and environmental quality. If some of these features are closely tied to crime, and if they can be changed by policy, it may be possible to lower crime without asking people to move. I ask three questions:

1. Is air quality associated with county crime rates?
2. Is cost of living associated with county crime rates?
3. Among the features available in the data, which are most strongly associated with crime once they are considered together?

1.2 Contribution and Scope

A first version of this project answered the first two questions with simple regressions on school-level records and found "highly significant" results for both. Working through the structure of the data showed that those tests were not valid: the school rows repeat the

same county and state values many times. This revised version moves the analysis to the county level, where crime is measured, uses standard errors clustered by state, and adds controls. The revision changes the conclusions, and Chapter 5 reports both the original and the corrected estimates so the difference is clear. The analysis is descriptive. It does not identify causal effects.

This edition, completed in October 2026, replaces the version submitted in December 2025 for the CDS 492 senior capstone course under the title "The Impact of Social and Economic Factors on Increasing Crime."

Chapter 2

Prior Research

2.1 Conceptual Framework

Dense areas put potential offenders closer to potential victims. From a rational-choice view, this lowers the cost of finding a target and raises the expected payoff of a crime. Density also changes how neighbors interact. Depending on how close a neighborhood is and how strong its institutions are, density can make informal social control stronger or weaker [1, 2].

Air and water pollution are a separate group of factors. Exposure may affect behavior through inflammation and reduced executive function [3, 4]. Some effects are short-run, such as more people and targets in a dense area. Others are long-run, such as childhood lead exposure. Community-level factors like collective efficacy sit between these two and can shape both.

2.2 Population Density and Crime Scaling

Large cities have higher crime rates than small towns and rural areas. Glaeser and Sacerdote [1] used victimization surveys, Uniform Crime Reports, and longitudinal data to argue that density raises the returns to crime. Urban scaling research finds that many social outcomes, including violent crime, grow faster than city population [7, 8, 9]. These scaling laws describe averages and do not predict any single city. At a finer scale, crime is concentrated in a small share of places even inside dense cities [10], so density alone does not explain crime rates. Homicide rates also differ a lot between countries with similar levels of urbanization, which points to governance and inequality as additional factors [6].

2.3 Collective Efficacy and Social Organization

Sampson, Raudenbush, and Earls [2] defined collective efficacy as mutual trust among neighbors plus a shared willingness to step in when something goes wrong. In their study of Chicago neighborhoods, higher collective efficacy went with lower violent crime, even after accounting for the neighborhood's structural characteristics. Neighborhood associations, religious groups, and community organizations can keep this kind of watchfulness going. When ties between neighbors are weak, high density can instead increase anonymity. Pollution and visible disorder may lower residents' willingness to act, so environmental quality and social organization can feed into each other [23].

2.4 Environmental Stress, Heat, and Aggression

Heat has been linked to aggression in experiments and in field data [12]. Studies using administrative records find more violent crime and conflict when temperatures are higher, after adjusting for seasonal patterns [13, 14, 15]. These studies fit the broader idea that environmental stress, from climate or from pollution, can change behavior.

2.5 Air Pollution and Violent Crime

Bondy, Roth, and Sager [3] used short-run changes in pollution and found that crime rises when air pollution rises, even after controlling for weather and time. A separate line of work shows that air pollution hurts cognitive performance [16]. That is consistent with a biological explanation, although the evidence about crime itself is thinner than the evidence about cognition.

2.6 Lead, Water Pollution, and Developmental Effects

Children exposed to lead show weaker impulse control and cognitive function. Reyes [4] used the phase-out of leaded gasoline and differences in exposure across birth cohorts, and linked lower childhood lead exposure to less violent crime later. The Flint, Michigan water crisis showed how failing infrastructure can raise blood lead levels in children in vulnerable communities [5, 18].

2.7 Land Use, Guardianship, and Policing

Routine activity theory says a crime happens when a motivated offender, a suitable target, and the lack of a capable guardian come together [11]. Land use touches all three. Areas that mix housing and commercial space can have more targets and more anonymity, while active businesses, residents, and community groups add guardianship. Hot-spot policing focuses on the places where crime clusters [19], and adding vegetation has been associated with less crime in inner-city neighborhoods [20].

2.8 Socioeconomic Inequality and Cumulative Environmental Risk

Inequality affects both who is exposed to environmental hazards and who is at risk of crime [21, 22, 17]. Disorder in a neighborhood can make residents feel less attached to it and weaken informal control [23]. The built environment is also linked to mental health [24]. For these reasons, pollution and crime should be looked at alongside broader disadvantage and not as separate problems.

2.9 Measurement, Identification, and Inference

Prior studies mostly use one of three designs: cross-sectional comparisons with many controls, panel models that use changes within a place over time, and quasi-experiments that rely on a policy change or an environmental shock [1, 3, 4]. Each has problems, such as spillovers between areas, crime moving somewhere else, and differences in how crime is reported.

A separate problem matters for this project. When a variable is measured for a large group (such as a state) but the regression is run on many smaller units inside each group (such as schools), the observations within a group share the same value and are not independent. Conventional standard errors then treat repeated copies as new information and can be far too small [25]. The usual fix is to cluster the standard errors at the level of the group [26]. My study is a cross-sectional comparison with controls and state-clustered standard errors. It can describe associations but not cause and effect.

Chapter 3

Data

3.1 Source

The data come from a public quality-of-life dataset posted on Kaggle [35], built to help people choose a place to live. Its compiler merged several U.S. government sources, and its source file lists them: reported crimes in 2016 from the FBI Uniform Crime Reporting county file [27]; 2022 county population estimates [28]; 2022 county unemployment from the USDA Economic Research Service [29]; family cost of living and median income from the Economic Policy Institute Family Budget Calculator [30]; the share of days with a "good" Air Quality Index from the EPA [31]; drinking-water violations from the EPA Safe Drinking Water Information System [32]; the 2020 presidential popular-vote winner by state [33]; and public school records from the National Center for Education Statistics [34]. The cost of living builds on an earlier Kaggle county dataset [36].

The dataset comes in two forms: a county file with one row per county (3,134 rows) and a school file with one row per public school (88,111 rows). The school file repeats its county's values on every school row.

3.2 Level of Measurement

Table 3.1 shows the level at which each variable actually varies. This is the key fact for the analysis. Crime, income, cost of living, unemployment, and population vary by county. Air quality and the vote winner vary only by state: the air-quality variable takes exactly one value per state and 50 distinct values in total. Any regression on air quality therefore has, in effect, about 50 independent observations of it, no matter how many rows are used.

Table 3.1: Variables and the level at which they vary.

Variable	Unit	Varies by	Year
Reported crime	per 1,000 residents	County	2016
Air quality	% of days with "good" AQI	State only	–
Drinking-water violations	count	County	–
Cost of living (family budget)	$ per year	County	–
Median household income	$ per year	County	2022
Cost-to-income ratio	%	County	–
Unemployment rate	%	County	2022
Population	persons	County	2022
Urban share of schools	share (0–1)	County (from school file)	2021–22
2020 popular-vote winner	D / R	State	2020

3.3 Construction of the Analysis Sample

I use the county file as the main data. Crime is stored as text such as "14/1000", which I convert to crimes per 1,000 residents. It is rounded to whole numbers. 120 counties, almost all very small, report a crime rate of exactly zero. A full year with no reported crime is more likely to mean that the county did not report to the FBI program than that no crime occurred, so I treat these as missing and check the result with them included.

The dataset has no land area, so I cannot compute population density directly. I measure urbanization in two ways instead: the log of county population, and the share of the county's public schools that the NCES classifies as being in a city or suburb. The school file has no Illinois schools, so Illinois counties have no urban share. They drop out of the main sample, and I check the result without the urban-share control so that they can be included.

After these steps the analysis sample has 2,892 counties in 49 states and the District of Columbia (50 state-level clusters; DC is a single county). Table 3.2 describes it.

Table 3.2: Descriptive statistics, analysis sample of counties.

Variable	N	Mean	SD	Min	Max
Crime per 1,000 residents (2016)	2,892	19.83	11.92	1.00	89.00
Air quality (% good days)	2,892	84.56	6.56	68.34	99.35
Cost of living ($)	2,892	70,869	10,500	54,431	167,022
Median income ($)	2,892	68,448	16,942	27,327	177,662
Cost-to-income ratio (%)	2,892	107.08	19.17	52.00	236.97
Unemployment rate (%)	2,892	3.72	1.32	1.34	17.19
Population (2022)	2,892	109,254	331,278	358	9,721,138
Urban share of schools	2,892	0.18	0.32	0.00	1.00
Drinking-water violations	2,620	2.96	11.05	0.00	456.00

3.4 Data Limitations

The variables come from different years. Crime is from 2016, while population, income, and unemployment are from 2022, so the explanatory variables are measured after the outcome (see Section 6.3). Air quality is a single state-level figure and cannot reflect differences between counties in the same state. Reported crime depends on what agencies report, and the crime measure is rounded to whole numbers per 1,000.

Chapter 4

Hypotheses and Methods

4.1 Hypotheses

Hypothesis 1 (air quality). H_0: air quality has no linear association with the county crime rate. H_1: it does. The literature in Chapter 2 suggests that better air quality should go with less crime.

Hypothesis 2 (cost of living). H_0: cost of living has no linear association with the county crime rate. H_1: it does.

4.2 Models

For county c in state s, the main specification is

$$\text{Crime}_{cs} = \beta_0 + \beta_1 \,\text{AirQuality}_s + \beta_2 \,\text{CostOfLiving}_{cs} + \mathbf{X}'_{cs}\boldsymbol{\gamma} + \varepsilon_{cs}, \tag{4.1}$$

where $\mathbf{X}$ holds the controls: log population, the urban share of schools, the unemployment rate, log median income, and the cost-to-income ratio. Because cost of living and median income are strongly correlated ($r = 0.72$), the full model uses the cost-to-income ratio, which measures how expensive a county is relative to what its residents earn. I estimate five models:

(1) air quality only;

(2) cost of living only;

(3) air quality and cost of living together;

(4) air quality with all controls;

(5) all controls plus state fixed effects.

Model (5) compares counties only with other counties in the same state. This removes everything that differs between states, including reporting practices, but it also removes air quality, which varies only by state. It also drops the District of Columbia and Utah, each of which has only one county in the sample, because a state with one county gives no within-state comparison. Model (5) therefore answers the third research question for the county-level variables.

All models are ordinary least squares with standard errors clustered by state, which allows counties in the same state to share unobserved shocks and accounts for the state-level air-quality measure. Because there are only 50 clusters, p-values use a t-distribution with 49 degrees of freedom. I use $\alpha = 0.05$. To compare variables measured in different units, I also report standardized coefficients: the change in crime, in standard deviations, for a one-standard-deviation change in each variable.

Because air quality varies only between states, it may simply pick up broad regional differences. As a direct check, I add fixed effects for the four Census regions (Northeast, Midwest, South, West) and, more finely, the nine Census divisions. These compare states with more and fewer good-air days within the same part of the country.

Chapter 5

Results

5.1 Why the First Version Overstated Significance

Table 5.1 starts with the original regression of crime on air quality, estimated on 87,457 school rows. The slope is the same in the first two lines; only the standard error changes. Treating the school rows as independent gives a t-statistic of -67. Clustering by state, which recognizes that air quality takes only one value per state, makes the standard error 15 times larger and the t-statistic falls to -4.5. The same slope estimated on counties, or on the 50 states directly, gives similar conclusions: the bivariate association is negative and significant, but far less precise than the first version reported. The original cost-of-living regression had the same problem, because it averaged school rows within ZIP codes and so repeated each county's values across its ZIP codes.

Table 5.1: Air quality and crime: the same bivariate slope under different units and standard errors. Slope is crimes per 1,000 residents per percentage point of good-air days.

Specification	Slope	SE	t	p
School rows, conventional SE (original)	-0.369	0.0055	-67.45	$< 10^{-16}$
School rows, SE clustered by state	-0.369	0.0824	-4.48	< 0.001
Counties, SE clustered by state	-0.303	0.0848	-3.57	< 0.001
States (n = 50), robust SE	-0.493	0.2023	-2.44	0.018

5.2 Main Estimates

Table 5.2 reports the county models.

Table 5.2: County regressions of crime per 1,000 residents. Standard errors clustered by state in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Model (5) drops the District of Columbia and Utah, which have one county each in the sample.

	(1)	(2)	(3)	(4)	(5)
Air quality (% good days)	−0.303***		−0.301***	−0.154**	
	(0.085)		(0.089)	(0.068)	
Cost of living ($10k)		−0.329	−0.153		
		(0.651)	(0.560)		
log population				3.086***	3.724***
				(0.458)	(0.635)
Urban share of schools				10.415***	9.867***
				(1.881)	(1.938)
Unemployment rate (%)				−0.202	0.590
				(0.473)	(0.388)
log median income				−27.781***	−32.434***
				(5.338)	(4.025)
Cost-to-income ratio (%)				−0.127***	−0.206***
				(0.046)	(0.041)
State fixed effects	No	No	No	No	Yes
Counties	2,892	2,892	2,892	2,892	2,890
States (clusters)	50	50	50	50	48
R^2	0.028	0.001	0.028	0.320	0.448

Air quality. On its own (Model 1), each additional percentage point of good-air days goes with 0.30 fewer crimes per 1,000 residents ($p < 0.001$). One standard deviation of air quality (6.6 points) therefore goes with about 2.0 fewer crimes per 1,000, or 10% of the mean of 19.8. Air quality alone explains only 2.8% of the variation between counties. Adding the controls (Model 4) cuts the coefficient in half, to −0.15 ($p = 0.028$). One standard deviation now goes with about 1.0 fewer crime per 1,000, or 5% of the mean. Figure 5.1 shows the same relationship across the 50 states, the level at which air quality is measured. Across states the bivariate slope is −0.49 ($p = 0.018$), but after controlling for state urbanization and income it is −0.42 with $p = 0.11$.

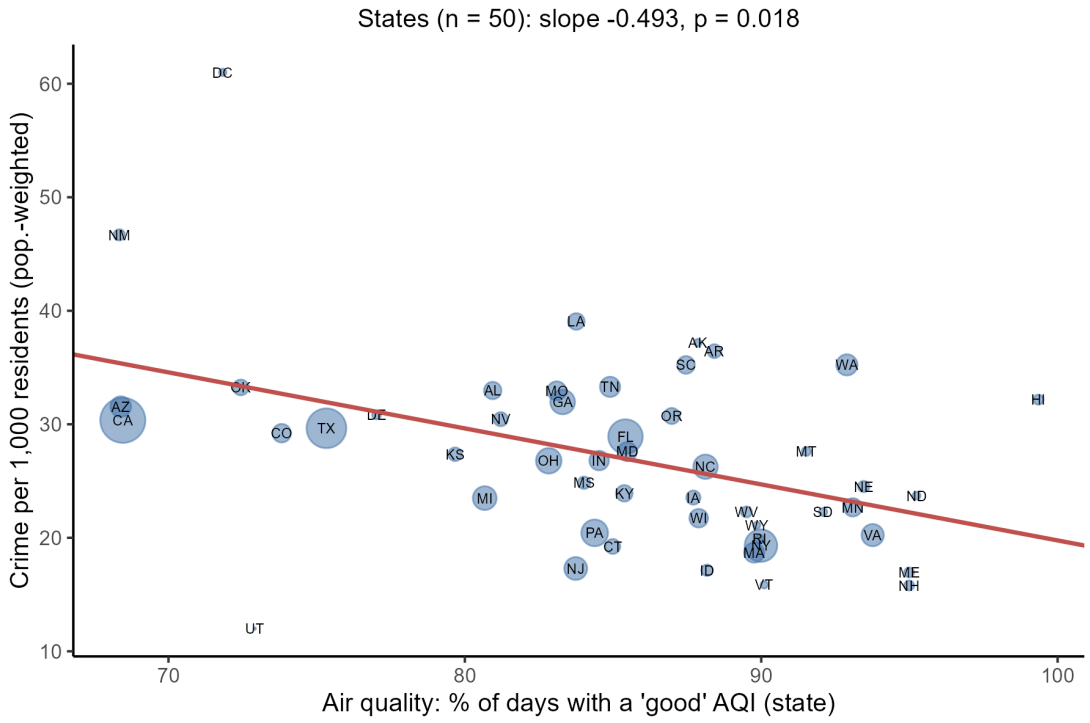


Figure 5.1: Crime and air quality across states. Each point is a state; crime is the population-weighted average over its counties, and point size reflects population. The line is the OLS fit.

Cost of living. Cost of living has no association with crime on its own (Model 2): the slope is -0.33 crimes per 1,000 per \$10,000, with a 95% confidence interval from -1.64 to 0.98 and $p = 0.62$. It also adds nothing when air quality is included (Model 3). Figure 5.2 shows why: across the range of cost of living, the average crime rate is nearly flat. The reason becomes clear in Model 4. Expensive counties tend to be both richer, which goes with less crime, and more urban, which goes with more crime, and the two cancel out. Holding income and urbanization fixed, a county that is more expensive relative to its income has somewhat less crime: a 10-point higher cost-to-income ratio goes with 1.3 fewer crimes per 1,000 ($p = 0.008$).

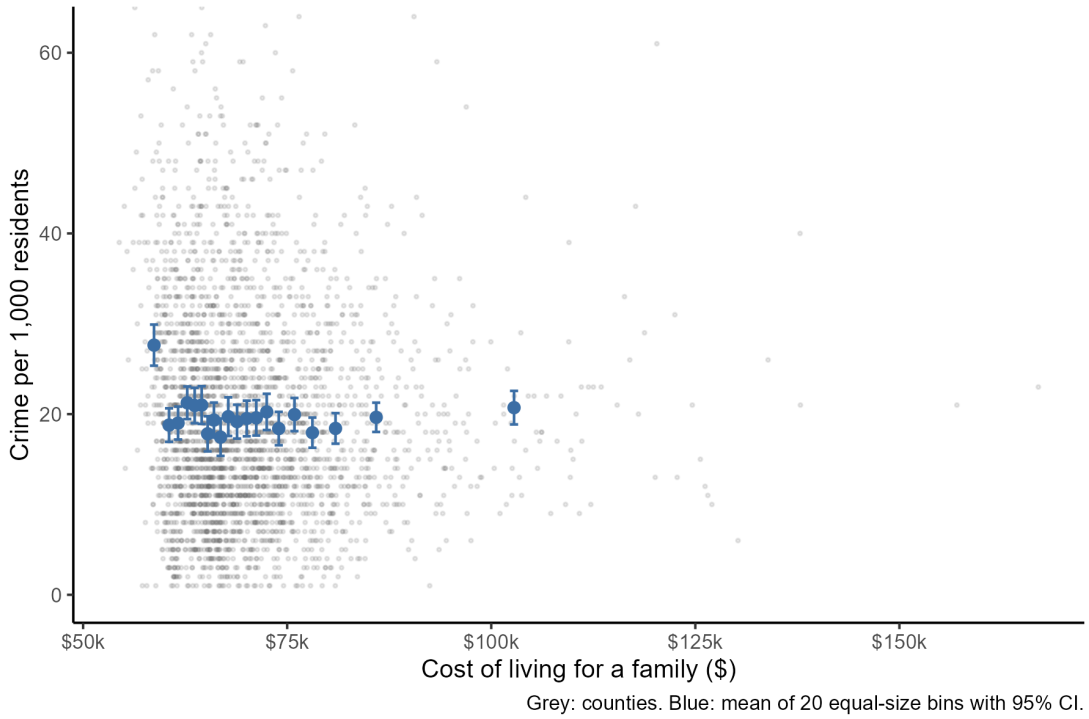


Figure 5.2: Crime and cost of living across counties. Grey points are counties; blue points are the mean crime rate in 20 equal-size groups of counties sorted by cost of living, with 95% confidence intervals.

Which features matter most. Figure 5.3 puts the variables on a common scale. Median income has the strongest association: one standard deviation higher income goes with 0.55 standard deviations less crime in Model 4 and 0.64 in Model 5. In crime units, a county with 10% higher median income has about 2.6 fewer crimes per 1,000. Next come population size (0.38 to 0.46 SD) and the urban share of schools (0.26 to 0.28 SD), both positive, and the cost-to-income ratio (-0.20 to -0.33 SD). Air quality is the smallest of the significant variables (-0.08 SD). Unemployment shows no association in either model. The controls raise the explained share of variation from 3% to 32%, and state fixed effects raise it to 45%.

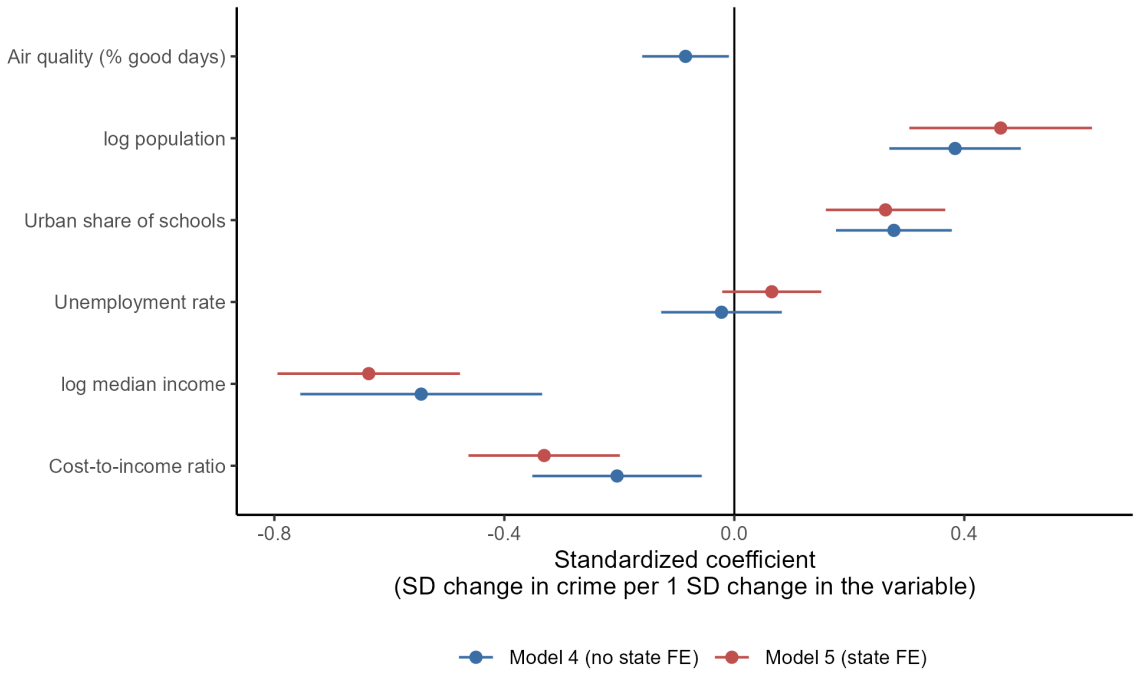


Figure 5.3: Standardized coefficients with 95% confidence intervals. Model 4 has no state fixed effects; Model 5 compares counties within the same state, so air quality (a state-level variable) drops out.

5.3 Robustness

Table 5.3 changes one choice at a time in Model 4 and reports the air-quality coefficient. Keeping the zero-crime counties, adding drinking-water violations, adding the state vote winner, using cost of living instead of the cost-to-income ratio, or dropping the urban-share control so that Illinois can be included all leave the coefficient between -0.14 and -0.19 with p between 0.020 and 0.047. Weighting counties by population, which gives large urban counties more say, keeps a similar coefficient (-0.19) but the standard error doubles and $p = 0.19$. Drinking-water violations themselves show no association with crime (coefficient 0.007, $p = 0.77$).

Air quality or geography? The last two rows of Table 5.3 compare counties only with other counties in the same Census region or division. With region fixed effects the air-quality coefficient falls from -0.15 to -0.06 ($p = 0.36$), and with division fixed effects to -0.05 ($p = 0.49$). This is not because the fixed effects remove most of the variation in air quality: region explains only 13% of the differences in air quality between states, and the standard error barely changes (0.068 to 0.069). The coefficient shrinks because the part

of air quality that is related to crime is the part shared within regions. Figure 5.1 shows the pattern: the states with the fewest good-air days are mostly in the West and Southwest (California, Arizona, New Mexico, Colorado, Utah), where ozone and dust are common. The same check at the state level gives the same answer (-0.32, $p = 0.12$, $n = 50$).

Table 5.3: Robustness of the air-quality coefficient in Model 4. Standard errors clustered by state.

Variation of Model (4)	Air quality coef. (SE)	p	Counties
Keep counties reporting zero crime	-0.165 (0.070)	0.023	3,004
Weight counties by population	-0.193 (0.144)	0.188	2,892
Add drinking-water violations	-0.156 (0.065)	0.020	2,620
Add state 2020 vote winner (R = 1)	-0.153 (0.074)	0.043	2,892
Cost of living instead of cost-to-income	-0.186 (0.077)	0.020	2,892
Drop urban share (adds Illinois)	-0.141 (0.069)	0.047	3,014
Add Census region fixed effects (4)	-0.063 (0.069)	0.365	2,892
Add Census division fixed effects (9)	-0.051 (0.072)	0.488	2,892

Chapter 6

Conclusion

6.1 Answers to the Research Questions

Q1: Is air quality associated with crime? Not in a reliable way. Counties in states with more good-air days have lower crime, and the main county model rejects H_0 at the 5% level. But the association is much smaller than the first version suggested, and it does not survive the checks. It halves once income and urbanization are taken into account, loses significance when counties are weighted by population or when states are the unit, and largely disappears when counties are compared within the same Census region or division. The pattern is best read as a regional difference, with the West having both fewer good-air days and somewhat higher crime, rather than as a link between air quality and crime. This does not contradict the pollution literature in Chapter 2, which finds effects of short-run changes in pollution on violent crime. A single state-level average of good-air days is too coarse to detect such effects.

Q2: Is cost of living associated with crime? No. I fail to reject H_0 ($p = 0.62$). The significant result in the first version was an artifact of repeating county values across school and ZIP-code rows. Cost of living matters only in relation to income: among counties with the same income and urbanization, those that are more expensive have somewhat less crime.

Q3: Which features matter most? Income and urbanization. Median income has the strongest association (negative), followed by population size and the urban share of schools (both positive). Air quality has the weakest association and is not robust to regional comparisons, and unemployment and drinking-water violations show none. These results match the older literature on density, opportunity, and disadvantage in Chapter 2 better than the newer environmental literature.

6.2 Lesson on Method

The largest change between the two versions came from the unit of analysis, not from a new model. A regression on 87,457 rows looked overwhelmingly significant, but the variable of interest had only 50 distinct values. Matching the unit of analysis to the level at which the data are measured, and clustering the standard errors at that level, is a basic step that changes the conclusions here. A second lesson is that a variable measured for large areas can easily stand in for geography. Comparing within regions is a simple way to check this.

6.3 Limitations

- **Associations only.** The data are a single cross-section. Unmeasured differences between places, such as policing, age structure, or poverty concentration, could explain the associations.
- **Coarse air-quality measure.** Air quality is one number per state, so the analysis cannot use differences in pollution between counties, and it has effectively 50 observations.
- **Reversed timing.** Crime is measured in 2016, but income, unemployment, and population are measured in 2022, six years later. The explanatory variables therefore come after the outcome they are meant to explain. This rules out reading them as causes of crime, and if crime itself affects later income or population, for example by pushing residents out, the associations could partly run in the opposite direction. The estimates assume that the ranking of counties on these variables changed little between 2016 and 2022.
- **Crime reporting.** The crime measure covers reported crimes only, is rounded to whole numbers per 1,000, and treats zero-crime counties as non-reporting.
- **No population density.** Without land area, urbanization is measured with population size and school locale codes. Illinois has no school records and is missing from the main sample.
- **Regional data.** Relationships across counties do not describe the behavior of individual people.

6.4 Future Work

The most useful next step is a county-level pollution measure, such as annual $PM_{2.5}$ from EPA monitors or satellite estimates, matched to crime in the same year. A panel of counties over several years would allow comparisons within the same county over time, which removes stable differences between places. Sudden pollution shocks, such as wildfire smoke, would come closer to the causal designs reviewed in Chapter 2. Adding land area, poverty, and age structure from the Census would also strengthen the county model.

Appendix A

Software

The first version of this project was written in R with `ggplot2`, `dplyr`, and `janitor`. The revised analysis is also written in R (version 4.3.2), using `dplyr` and `readr` for data preparation, `fixest` for the regressions and clustered standard errors, `ggplot2` for the figures, and `jsonlite` to save the numbers quoted in the text. It reads the county and school files, builds the analysis sample, estimates all models, and writes every table and figure in this document. The full script is below.

```
# Undergraduate thesis, CDS 492 (revised analysis): county-level crime, air quality, cost of living
# Austin Yoo
#
# Re-estimates the thesis models at the county level, where crime is measured,
# with state-clustered standard errors and controls. Writes LaTeX tables, figures,
# and a JSON of key numbers used in the paper.
#
# Usage: Rscript analysis.R   (run from this folder)

suppressPackageStartupMessages(library(dplyr))
suppressPackageStartupMessages(library(readr))
suppressPackageStartupMessages(library(ggplot2))
suppressPackageStartupMessages(library(fixest))
suppressPackageStartupMessages(library(jsonlite))

# Raw data are not redistributed here. Download them from Kaggle into data/
# (see data/README.md): QOL(County Level).csv and QOL(Public School Level).csv.
DATA <- "data"
OUT  <- "output"
FIG  <- "paper"
dir.create(OUT, showWarnings = FALSE)
dir.create(FIG, showWarnings = FALSE)

# ---------------------------------------------------------------------------
# Cleaning helpers
# ---------------------------------------------------------------------------
# '$75,531.37' / '3.21%' / '6,605' -> numeric. '-1' and blanks -> NA.
num <- function(s) {
```

```r
  x <- suppressWarnings(as.numeric(gsub("[$,%]", "", trimws(s))))
  ifelse(!is.na(x) & x == -1, NA, x)
}

# '  8/1000' -> 8 crimes per 1,000 residents. '0' -> NA (treated as non-reporting).
crime_per_1k <- function(s) {
  s <- trimws(s)
  parts <- strsplit(s, "/", fixed = TRUE)
  sapply(parts, function(v) if (length(v) == 2) as.numeric(v[1]) / as.numeric(v[2]) * 1000 else NA_real_)
}

crime_per_1k_keep_zero <- function(s) {
  r <- crime_per_1k(s)
  ifelse(trimws(s) == "0", 0, r)
}

# ------------------------------------------------------------------------
# County-level data
# ------------------------------------------------------------------------
c_raw <- read_csv(file.path(DATA, "QOL(County Level).csv"),
                  col_types = cols(.default = col_character()))
county <- tibble(
  fips     = sprintf("%05d", as.integer(c_raw$FIPS)),
  state    = c_raw$LSTATE,
  crime    = crime_per_1k(c_raw$`2016 Crime Rate`),
  crime0   = crime_per_1k_keep_zero(c_raw$`2016 Crime Rate`),
  pop      = num(c_raw$`2022 Population`),
  unemp    = num(c_raw$Unemployment),
  aqi_good = num(c_raw$`AQI%Good`),
  water_vpv = num(c_raw$WaterQualityVPV),
  col      = num(c_raw$`Cost of Living`),
  income   = num(c_raw$`2022 Median Income`),
  c2i      = num(c_raw$`AVG C2I`),
  party_r  = as.integer(trimws(c_raw$`2020PopulrVoteParty`) == "R")
)

# Urbanicity from NCES school locale codes: share of a county's public schools
# located in a City (11-13) or Suburb (21-23) locale.
s <- read_csv(file.path(DATA, "QOL(Public School Level).csv"),
              col_types = cols(.default = col_character()),
              col_select = c(FIPS, ULOCALE, LSTATE, `2016 Crime Rate`, `AQI%Good`),
              locale = locale(encoding = "latin1"), name_repair = "unique_quiet")
s <- s %>% mutate(fips = sprintf("%05d", as.integer(FIPS)),
                  urban = as.numeric(substr(ULOCALE, 1, 1) %in% c("1", "2")))
loc <- s %>% group_by(fips) %>%
  summarize(urban_share = mean(urban), n_schools = n(), .groups = "drop")
county <- county %>% left_join(loc, by = "fips") %>%
  mutate(log_pop = log(pop), log_income = log(income), col_10k = col / 10000)

# Census regions and divisions, used to check whether air quality only proxies for
# broad geography (e.g., the dry, sunny West).
divisions <- list(
  "New England" = c("CT", "ME", "MA", "NH", "RI", "VT"),
  "Middle Atlantic" = c("NJ", "NY", "PA"),
  "East North Central" = c("IL", "IN", "MI", "OH", "WI"),
```

```
  "West North Central" = c("IA", "KS", "MN", "MO", "NE", "ND", "SD"),
  "South Atlantic" = c("DE", "DC", "FL", "GA", "MD", "NC", "SC", "VA", "WV"),
  "East South Central" = c("AL", "KY", "MS", "TN"),
  "West South Central" = c("AR", "LA", "OK", "TX"),
  "Mountain" = c("AZ", "CO", "ID", "MT", "NV", "NM", "UT", "WY"),
  "Pacific" = c("AK", "CA", "HI", "OR", "WA"))
div_region <- c("New England" = "Northeast", "Middle Atlantic" = "Northeast",
                "East North Central" = "Midwest", "West North Central" = "Midwest",
                "South Atlantic" = "South", "East South Central" = "South",
                "West South Central" = "South", "Mountain" = "West", "Pacific" = "West")
state_div <- setNames(rep(names(divisions), lengths(divisions)), unlist(divisions))
county <- county %>% mutate(division = unname(state_div[state]),
                            region = unname(div_region[division]))
stopifnot(!anyNA(county$region))

# ------------------------------------------------------------------------------
# Structure checks reported in the paper
# ------------------------------------------------------------------------------
facts <- list(
  n_counties_file = nrow(county),
  n_states = n_distinct(county$state),
  aqi_unique_values = n_distinct(county$aqi_good, na.rm = TRUE),
  aqi_max_unique_within_state = max(tapply(county$aqi_good, county$state,
                                           function(x) n_distinct(x, na.rm = TRUE))),
  crime_zero_counties = sum(trimws(c_raw$`2016 Crime Rate`) == "0"),
  n_school_rows = nrow(s),
  states_missing_from_school_file = setdiff(unique(county$state), unique(s$LSTATE))
)

# ------------------------------------------------------------------------------
# Model 0: replicate the original school-level regression (n ~ 87k)
# ------------------------------------------------------------------------------
s0 <- s %>% mutate(crime_rate = crime_per_1k(`2016 Crime Rate`) / 1000,  # original scale
                   aqi_good = num(`AQI%Good`)) %>%
  filter(!is.na(crime_rate), !is.na(aqi_good))
m0  <- feols(crime_rate ~ aqi_good, data = s0, vcov = "iid")
m0c <- feols(crime_rate ~ aqi_good, data = s0, cluster = ~LSTATE)
facts$orig <- list(
  n = nobs(m0), b = unname(coef(m0)["aqi_good"]),
  se_iid = unname(se(m0)["aqi_good"]), se_cluster = unname(se(m0c)["aqi_good"]),
  t_iid = unname(tstat(m0)["aqi_good"]), t_cluster = unname(tstat(m0c)["aqi_good"]),
  p_cluster = unname(pvalue(m0c)["aqi_good"]), r2 = unname(r2(m0, "r2")),
  se_ratio = unname(se(m0c)["aqi_good"] / se(m0)["aqi_good"])
)

# ------------------------------------------------------------------------------
# County models (OLS, standard errors clustered by state)
# ------------------------------------------------------------------------------
need <- c("crime", "aqi_good", "log_pop", "urban_share", "unemp", "log_income", "c2i", "col_10k")
d <- county %>% filter(if_all(all_of(need), ~ !is.na(.x)))
facts$n_analysis <- nrow(d)
facts$n_states_analysis <- n_distinct(d$state)
facts$singleton_states <- names(which(table(d$state) == 1))

controls <- "log_pop + urban_share + unemp + log_income + c2i"
```

```
fit <- function(f, data, fe = FALSE, ...) {
  f <- if (fe) paste(f, "| state") else f
  feols(as.formula(f), data = data, cluster = ~state, ...)
}

models <- list(
  M1 = fit("crime ~ aqi_good", d),
  M2 = fit("crime ~ col_10k", d),
  M3 = fit("crime ~ aqi_good + col_10k", d),
  M4 = fit(paste("crime ~ aqi_good +", controls), d),
  M5 = fit(paste("crime ~", controls), d, fe = TRUE)
)

# State-level regression (the level at which air quality actually varies)
st <- d %>% group_by(state) %>%
  summarize(crime = weighted.mean(crime, pop), aqi_good = first(aqi_good),
            urban_share = weighted.mean(urban_share, pop),
            log_income = weighted.mean(log_income, pop), pop = sum(pop),
            region = first(region), .groups = "drop")
ms1 <- feols(crime ~ aqi_good, data = st, vcov = "hetero")
ms2 <- feols(crime ~ aqi_good + urban_share + log_income, data = st, vcov = "hetero")
ms3 <- feols(crime ~ aqi_good | region, data = st, vcov = "hetero")

# Standardized coefficients (1 SD change in X -> SD change in crime)
labels <- c(aqi_good = "Air quality (% good days)", log_pop = "log population",
            urban_share = "Urban share of schools", unemp = "Unemployment rate",
            log_income = "log median income", c2i = "Cost-to-income ratio")
sd_y <- sd(d$crime)
std <- bind_rows(lapply(names(labels), function(v) {
  bind_rows(lapply(c("M4", "M5"), function(mk) {
    m <- models[[mk]]
    if (!v %in% names(coef(m))) return(NULL)
    k <- sd(d[[v]]) / sd_y
    ci <- confint(m)[v, ]
    tibble(var = v, label = labels[[v]], model = mk, beta = coef(m)[[v]] * k,
           lo = ci[[1]] * k, hi = ci[[2]] * k, p = pvalue(m)[[v]])
  }))
}))
write_csv(std, file.path(OUT, "standardized_coefs.csv"))

# ---------------------------------------------------------------------------
# Robustness: one change at a time to Model 4
# ---------------------------------------------------------------------------
f4 <- paste("crime ~ aqi_good +", controls)
d0 <- county %>% filter(if_all(all_of(c("crime0", "aqi_good", "log_pop", "urban_share",
                                       "unemp", "log_income", "c2i")), ~ !is.na(.x))) %>%
  mutate(crime = crime0)
# Illinois has no rows in the school file, so it has no urban share. Dropping that
# control brings Illinois back into the sample.
dil <- county %>% filter(if_all(all_of(c("crime", "aqi_good", "log_pop", "unemp",
                                        "log_income", "c2i")), ~ !is.na(.x)))
rob <- list(
  keep_zero        = fit(f4, d0),
  pop_weighted     = fit(f4, d, weights = ~pop),
  water            = fit(paste(f4, "+ water_vpv"), filter(d, !is.na(water_vpv))),
```

```
  party            = fit(paste(f4, "+ party_r"), d),
  col_instead_c2i  = fit("crime ~ aqi_good + log_pop + urban_share + unemp + col_10k", d),
  no_urban_with_il = fit("crime ~ aqi_good + log_pop + unemp + log_income + c2i", dil),
  region_fe        = fit(paste(f4, "| region"), d),
  division_fe      = fit(paste(f4, "| division"), d)
)

# How much of the air-quality variation is left within regions and divisions
facts$aqi_sd_states <- sd(st$aqi_good)
facts$aqi_sd_within_region <- sd(st$aqi_good - ave(st$aqi_good, st$region))
facts$aqi_r2_on_region <- summary(lm(aqi_good ~ region, data = st))$r.squared

# ---------------------------------------------------------------------------
# LaTeX tables
# ---------------------------------------------------------------------------
f3 <- function(x, k = 3) formatC(x, format = "f", digits = k)
pfmt <- function(p) if (p < 0.001) "<0.001" else f3(p)
stars <- function(p) if (p < 0.01) "^{***}" else if (p < 0.05) "^{**}" else if (p < 0.1) "^{*}" else ""
cell <- function(m, v) {
  if (!v %in% names(coef(m))) return(c("", ""))
  c(sprintf("$%s%s$", f3(coef(m)[[v]]), stars(pvalue(m)[[v]])), sprintf("$(%s)$", f3(se(m)[[v]])))
}

rows <- c(aqi_good = "Air quality (\\% good days)", col_10k = "Cost of living (\\$10k)",
          log_pop = "log population", urban_share = "Urban share of schools",
          unemp = "Unemployment rate (\\%)", log_income = "log median income",
          c2i = "Cost-to-income ratio (\\%)")
ord <- names(models)
L <- c(paste0("\\begin{tabular}{@{}l", strrep("c", length(ord)), "@{}}"), "\\toprule",
       paste0(" & ", paste0("(", seq_along(ord), ")", collapse = " & "), " \\\\"), "\\midrule")
for (v in names(rows)) {
  a <- lapply(models, cell, v = v)
  L <- c(L, paste0(rows[[v]], " & ", paste(sapply(a, `[`, 1), collapse = " & "), " \\\\"),
         paste0(" & ", paste(sapply(a, `[`, 2), collapse = " & "), " \\\\[2pt]"))
}
L <- c(L, "\\midrule",
       paste0("State fixed effects & ", paste(ifelse(ord == "M5", "Yes", "No"), collapse = " & "), " \\\\"),
       paste0("Counties & ", paste(sapply(models, function(m) format(nobs(m), big.mark = ",")), collapse = " & "), " \\\\"),
       paste0("States (clusters) & ", paste(sapply(models, function(m) n_distinct(d$state[obs(m)])), collapse = " & "), " \\\\"),
       paste0("$R^2$ & ", paste(sapply(models, function(m) f3(r2(m, "r2"))), collapse = " & "), " \\\\"),
       "\\bottomrule", "\\end{tabular}")
writeLines(L, file.path(OUT, "table_main.tex"))

# Original vs corrected inference for the air-quality slope
o <- facts$orig
m1 <- models$M1
row_inf <- function(lab, b, s, t, p) sprintf("%s & $%s$ & $%s$ & $%s$ & $%s$ \\\\", lab, f3(b), f3(s, 4), f3(t, 2), p)
L <- c("\\begin{tabular}{@{}lcccc@{}}", "\\toprule",
       "Specification & Slope & SE & $t$ & $p$ \\\\", "\\midrule",
       row_inf("School rows, conventional SE (original)", o$b * 1000, o$se_iid * 1000, o$t_iid, "<10^{-16}"),
       row_inf("School rows, SE clustered by state", o$b * 1000, o$se_cluster * 1000, o$t_cluster, pfmt(o$p_cluster)),
```

```
       row_inf("Counties, SE clustered by state", coef(m1)[["aqi_good"]], se(m1)[["aqi_good"]],
               tstat(m1)[["aqi_good"]], pfmt(pvalue(m1)[["aqi_good"]])),
       row_inf(sprintf("States (n = %d), robust SE", nobs(ms1)), coef(ms1)[["aqi_good"]], se(ms1)[["aqi_good"]],
               tstat(ms1)[["aqi_good"]], pfmt(pvalue(ms1)[["aqi_good"]])),
       "\\bottomrule", "\\end{tabular}")
writeLines(L, file.path(OUT, "table_inference.tex"))

# Robustness table: air-quality coefficient and its p-value
rob_lab <- c(keep_zero = "Keep counties reporting zero crime",
             pop_weighted = "Weight counties by population",
             water = "Add drinking-water violations",
             party = "Add state 2020 vote winner (R = 1)",
             col_instead_c2i = "Cost of living instead of cost-to-income",
             no_urban_with_il = "Drop urban share (adds Illinois)",
             region_fe = "Add Census region fixed effects (4)",
             division_fe = "Add Census division fixed effects (9)")
L <- c("\\begin{tabular}{@{}lccc@{}}", "\\toprule",
       "Variation of Model (4) & Air quality coef. (SE) & $p$ & Counties \\\\", "\\midrule")
for (k in names(rob_lab)) {
  m <- rob[[k]]
  L <- c(L, sprintf("%s & $%s$ $(%s)$ & $%s$ & %s \\\\", rob_lab[[k]], f3(coef(m)[["aqi_good"]]),
                    f3(se(m)[["aqi_good"]]), pfmt(pvalue(m)[["aqi_good"]]), format(nobs(m), big.mark = ",")))
}
L <- c(L, "\\bottomrule", "\\end{tabular}")
writeLines(L, file.path(OUT, "table_robust.tex"))

# Descriptive statistics
desc_vars <- c(crime = "Crime per 1,000 residents (2016)", aqi_good = "Air quality (\\% good days)",
               col = "Cost of living (\\$)", income = "Median income (\\$)",
               c2i = "Cost-to-income ratio (\\%)", unemp = "Unemployment rate (\\%)",
               pop = "Population (2022)", urban_share = "Urban share of schools",
               water_vpv = "Drinking-water violations")
L <- c("\\begin{tabular}{@{}lrrrrr@{}}", "\\toprule",
       "Variable & N & Mean & SD & Min & Max \\\\", "\\midrule")
for (v in names(desc_vars)) {
  x <- d[[v]][!is.na(d[[v]])]
  k <- if (max(abs(x)) > 1000) 0 else 2
  fm <- function(z) formatC(z, format = "f", digits = k, big.mark = ",")
  L <- c(L, sprintf("%s & %s & %s & %s & %s & %s \\\\", desc_vars[[v]], format(length(x), big.mark = ","),
                    fm(mean(x)), fm(sd(x)), fm(min(x)), fm(max(x))))
}
L <- c(L, "\\bottomrule", "\\end{tabular}")
writeLines(L, file.path(OUT, "table_desc.tex"))

# ---------------------------------------------------------------------------
# Figures
# ---------------------------------------------------------------------------
theme_set(theme_classic(base_size = 10))

# Fig 1: state-level scatter, crime vs air quality
g1 <- ggplot(st, aes(aqi_good, crime)) +
  geom_point(aes(size = pop), alpha = 0.5, color = "#3b6ea5", show.legend = FALSE) +
  geom_text(aes(label = state), size = 2) +
  geom_abline(intercept = coef(ms1)[[1]], slope = coef(ms1)[[2]], color = "#c0504d", linewidth = 0.8) +
```

```
  scale_size_area(max_size = 8) +
  labs(x = "Air quality: % of days with a 'good' AQI (state)",
       y = "Crime per 1,000 residents (pop.-weighted)",
       title = sprintf("States (n = %d): slope %s, p = %s", nobs(ms1),
                       f3(coef(ms1)[[2]]), f3(pvalue(ms1)[[2]]))) +
  theme(plot.title = element_text(size = 10, hjust = 0.5))
ggsave(file.path(FIG, "fig_state_aqi.png"), g1, width = 6, height = 4, dpi = 300)

# Fig 2: county binned scatter, crime vs cost of living (20 equal-count bins)
bins <- d %>% mutate(bin = ntile(col, 20)) %>% group_by(bin) %>%
  summarize(col = mean(col), se = sd(crime) / sqrt(n()), crime = mean(crime), .groups = "drop")
g2 <- ggplot() +
  geom_point(data = d, aes(col, crime), size = 0.4, alpha = 0.15, color = "grey40") +
  geom_errorbar(data = bins, aes(col, ymin = crime - 1.96 * se, ymax = crime + 1.96 * se),
                width = 1000, color = "#3b6ea5") +
  geom_point(data = bins, aes(col, crime), color = "#3b6ea5", size = 1.8) +
  scale_x_continuous(labels = function(v) paste0("$", v / 1000, "k")) +
  coord_cartesian(ylim = c(0, quantile(d$crime, 0.995))) +
  labs(x = "Cost of living for a family ($)", y = "Crime per 1,000 residents",
       caption = "Grey: counties. Blue: mean of 20 equal-size bins with 95% CI.")
ggsave(file.path(FIG, "fig_county_col.png"), g2, width = 6, height = 4, dpi = 300)

# Fig 3: standardized coefficients
std_plot <- std %>% mutate(label = factor(label, levels = rev(unname(labels))),
                           model = recode(model, M4 = "Model 4 (no state FE)", M5 = "Model 5 (state FE)"))
g3 <- ggplot(std_plot, aes(beta, label, color = model)) +
  geom_vline(xintercept = 0, linewidth = 0.4) +
  geom_pointrange(aes(xmin = lo, xmax = hi), position = position_dodge(width = 0.5), size = 0.3) +
  scale_color_manual(values = c("#3b6ea5", "#c0504d")) +
  labs(x = "Standardized coefficient\n(SD change in crime per 1 SD change in the variable)",
       y = NULL, color = NULL) +
  theme(legend.position = "bottom")
ggsave(file.path(FIG, "fig_std_coefs.png"), g3, width = 6.5, height = 4, dpi = 300)

# ---------------------------------------------------------------------------
# Numbers used in the text
# ---------------------------------------------------------------------------
pack <- function(m, v) {
  ci <- confint(m)[v, ]
  list(b = coef(m)[[v]], se = se(m)[[v]], p = pvalue(m)[[v]], lo = ci[[1]], hi = ci[[2]])
}
facts$mean_crime <- mean(d$crime)
facts$sd_crime <- sd(d$crime)
facts$sd_aqi <- sd(d$aqi_good)
facts$corr_col_income <- cor(d$col, d$income)
for (k in names(models)) {
  m <- models[[k]]
  facts[[k]] <- c(list(n = nobs(m), r2 = unname(r2(m, "r2"))),
                  lapply(setNames(nm = intersect(names(rows), names(coef(m)))), function(v) pack(m, v)))
}
facts$state_M1 <- c(list(n = nobs(ms1), r2 = unname(r2(ms1, "r2"))), pack(ms1, "aqi_good"))
facts$state_M2 <- c(list(n = nobs(ms2), r2 = unname(r2(ms2, "r2"))), pack(ms2, "aqi_good"))
facts$state_M3_region <- c(list(n = nobs(ms3), r2 = unname(r2(ms3, "r2"))), pack(ms3, "aqi_good"))
facts$robust <- lapply(rob, function(m) c(list(n = nobs(m)), pack(m, "aqi_good")))
facts$robust$water_coef <- pack(rob$water, "water_vpv")
```

```
facts$std <- std
write_json(facts, file.path(OUT, "facts.json"), auto_unbox = TRUE, digits = NA, pretty = TRUE)

etable(models, cluster = ~state)
print(summary(ms1)); print(summary(ms2))
print(std, n = Inf)
str(facts[c("n_analysis", "n_states_analysis", "states_missing_from_school_file", "orig")])
```

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Biography

Austin Yoo is completing a Bachelor of Science in Computational and Data Sciences at George Mason University, with a minor in Business and an expected graduation in May 2026. He is interested in environmental data, social behavior, decision making, simulation modeling, and machine learning applied to social and economic systems.

During his studies he worked on projects about crime prediction, housing market trends, and social behavior simulations using Python, R, and NetLogo. His business minor coursework covered business analytics, management, and IT strategy.

After graduation he plans to keep doing research on social, economic, and financial behavior with data science, mainly analytical modeling and financial technology.